

Zero-Shot Adapter Routing via Instruction Prefix Embeddings

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Abstract

Multi-task language models benefit from specialized LoRA adapters, but selecting the appropriate adapter at inference typically requires validation examples that are unavailable in zero-shot settings. This work introduces Instruction-Prefix-Conditioned Routing (IPCR), which routes instructions to adapters using frozen sentence embeddings and a linear probe without any task-specific validation data. The key finding is that instruction prefixes encoded by MiniLM-L6-v2 are linearly separable by task family (macro-F1 = 0.995), enabling a linear probe to achieve 72.67% top-1 accuracy (95.78% top-3) in predicting the oracle adapter. Zero-shot IPCR routing achieves 95.00% of oracle adapter performance on FLAN task families. However, robustness experiments reveal a limitation: routing depends on lexical keywords rather than semantic invariants, with 44.4% accuracy drops under 50% keyword masking and cosine similarity of 0.782 under paraphrase perturbations. IPCR is suitable for controlled instruction formats but requires robustification for open-ended queries.

1. Introduction

Adapter selection in multi-task models typically requires validation data that is unavailable in zero-shot deployment scenarios. Current routing methods address this through various mechanisms: LoRAHub uses gradient-free optimization over validation loss; LORAUTER routes via task embeddings computed from validation examples; MoELoRA trains routing jointly with adapters. All assume access to task-

specific validation data—at minimum 5 examples per task—to make routing decisions.

A fundamental question remains unaddressed: does the instruction itself contain sufficient signal for adapter routing, without any task examples? This work hypothesizes that the alignment between instruction semantics and adapter specializations exists intrinsically, inherited from the base model’s instruction-tuning.

The central finding is that instruction prefixes embedded by frozen sentence encoders are linearly separable by task family. Using MiniLM-L6-v2 embeddings of FLAN instruction prefixes, a logistic regression classifier achieves macro-F1 = 0.995 on task family classification. This separability enables a linear probe to predict the optimal adapter with 72.67% top-1 accuracy (95.78% top-3), and the resulting zero-shot routing achieves 95.00% of oracle adapter performance.

This work makes three contributions:

- 1. Empirical demonstration of intrinsic alignment.** Instruction embeddings from a frozen sentence encoder cluster by task family with near-perfect linear separability (F1 = 0.995), indicating that instruction semantics and adapter specializations share geometric structure without requiring joint training.
- 1. Zero-shot adapter routing (IPCR).** A method using frozen MiniLM embeddings and a linear probe that achieves 95.00% of oracle performance on FLAN task families, eliminating the need for validation data.
- 1. Characterization of routing fragility.** Routing depends on lexical keywords rather than semantic invariants. Paraphrase perturbations cause embedding drift (cosine similarity 0.782), and 50% keyword masking produces 44.4% accuracy drops.

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2. Related Work

2.0.1. PARAMETER-EFFICIENT FINE-TUNING

Low-Rank Adaptation (LoRA) introduced rank-decomposition matrices injected into transformer layers, achieving comparable performance to full fine-tuning with less than 1% trainable parameters. AdapterFusion demonstrated that task-specific adapters can be combined non-destructively through learned attention, outperforming multi-task learning on 16 NLU tasks. These methods establish that adapter combination is both mathematically valid (LoRA deltas are additive) and empirically effective, but require knowing which adapters to combine for a given input.

2.0.2. DYNAMIC ADAPTER ROUTING

LoRAHub achieves cross-task generalization via gradient-free composition, optimizing adapter weights over few-shot validation loss. The approach achieves upper-bound performance exceeding in-context learning on Big-Bench Hard, but requires validation examples to compute the optimization objective.

LORAUTER routes queries using task embeddings computed from validation sets, achieving 101.2% of oracle performance on task-aligned adapters and +5.2 points on unseen tasks. The method scales to 1500+ adapters but relies on 5+ validation examples per task to construct the routing representation.

MoELoRA treats LoRA modules as MoE experts with contrastive learning to encourage distinct specialization, achieving +4.2% over vanilla LoRA on math reasoning. However, routing is jointly trained with adapters, precluding generalization to new tasks without retraining.

Multi-Head Adapter Routing (MHR) combines adapter subsets with learned routing weights. HMoRA extends this to hierarchical routing with auxiliary losses.

All existing routing methods either require validation data, joint training, or gradient signals at test time. None test whether instruction-adapter alignment exists intrinsically without any task-specific training signal.

2.0.3. INSTRUCTION TUNING AND TASK REPRESENTATIONS

FLAN demonstrated that instruction-tuned models condition behavior on instruction format across diverse task families. T0 showed zero-shot generalization through unified prompting. These works suggest that

instruction prefixes encode meaningful task semantics.

Sentence encoders like MiniLM and Sentence-BERT compress text into fixed-dimensional embeddings via contrastive pretraining. SimCSE demonstrated that such embeddings provide paraphrase invariance—a property tested explicitly in the robustness experiments.

3. Methodology

3.0.1. OVERVIEW

Given an instruction x , IPCR routes to a bank of k task-specific LoRA adapters through three steps:

1. **Embed:** Encode the instruction prefix using a frozen sentence encoder $f : X \rightarrow \mathbb{R}^d$
2. **Route:** Apply a linear probe $W \in \mathbb{R}^{k \times d}$ to predict adapter scores
3. **Select:** Choose the highest-scoring adapter (hard routing)

If instruction semantics and adapter specializations share geometric structure, a linear mapping should suffice. Any success demonstrates intrinsic alignment, and failure would motivate more complex architectures.

3.0.2. INSTRUCTION EMBEDDING

MiniLM-L6-v2, a 22M-parameter sentence encoder pretrained via contrastive learning, serves as the embedding function. The encoder is completely frozen during both probe training and inference. Only the instruction prefix (first 128 tokens) is encoded, excluding input/output content. This focuses routing on task semantics rather than instance-specific features.

3.0.3. LINEAR PROBE TRAINING

The routing probe is a multinomial logistic regression classifier:

$$p(y = j \mid x) = \exp(W_j \cdot f(x)) / \sum_i \exp(W_i \cdot f(x))$$

where W_j is the weight vector for adapter j . Training uses L-BFGS optimization with balanced class weights to handle task-frequency imbalance in FLAN.

A linear probe has minimal capacity—it can only exploit structure that already exists in the embedding space. If the probe succeeds, this provides evidence that MiniLM embeddings encode task-relevant structure.

3.0.4. ADAPTER BANK

Each adapter is a task-specialized LoRA with rank $r = 16$, scaling factor $\alpha = 32$, applied to query/value projections in all attention layers. Adapters are trained independently on their respective task families, then frozen for routing experiments.

3.0.5. BASELINE COMPARISONS

To isolate routing contribution, comparisons include:

- **Oracle:** Always select the adapter trained on the input’s task family (upper bound)
- **Uniform:** Equal weight to all adapters (no routing information)
- **Random:** Uniformly random adapter selection

4. Experimental Setup

4.0.1. RESEARCH QUESTIONS

- **RQ1 (H-E0):** Are instruction prefix embeddings linearly separable by FLAN task family?
- **RQ2 (H-E1):** Can a linear probe predict the oracle adapter selection with high accuracy?
- **RQ3 (H-M1):** Does zero-shot IPCR routing achieve $\geq 90\%$ of oracle adapter performance?
- **RQ4 (H-M2):** Is routing robust to paraphrase and keyword perturbations?

4.0.2. DATASET

Evaluation uses the FLAN instruction-tuning collection accessed via the Open-Orca/FLAN HuggingFace repository.

Property	Value
Total samples	50,000 (streaming subset)
Task families	9-18 (depending on experiment)
Embedding dimension	384 (MiniLM-L6-v2)
Train/Test split	70/15/15 (stratified)

Task families tested include: `cot_gsm8k` (math), `cot_strategyqa` (reasoning), `cot_creak` (verification), `cot_qasc` (science), `cot_ecqa` (common-sense), `cot_sensemaking` (logic), `cot_esnli` (NLI), `cot_aqua_rat` (algebra), `stream_qed`, and others.

4.0.3. IMPLEMENTATION DETAILS

Encoder: sentence-transformers/all-MiniLM-L6-v2 (frozen, 22M parameters)

Classifier: Scikit-learn LogisticRegression with L-BFGS solver, max iterations 2000, balanced class weights

Adapter configuration:

- Base model: Mistral-7B-Instruct-v0.1
- LoRA rank: 16
- LoRA alpha: 32
- Target modules: `q_proj`, `v_proj`

Compute: Single NVIDIA A100 (40GB)

4.0.4. EVALUATION METRICS

For separability (RQ1): Macro-F1 score (threshold: ≥ 0.75), per-family F1 scores

For adapter selection (RQ2): Top-1 accuracy (threshold: $\geq 70\%$), Top-3 accuracy (threshold: $\geq 85\%$)

For end-to-end performance (RQ3): Relative performance (IPCR accuracy / Oracle accuracy), threshold: $\geq 90\%$ of oracle

For robustness (RQ4): Cosine similarity under paraphrase (threshold: ≥ 0.90), accuracy drop under keyword masking (threshold: $<10\%$)

5. Results

5.0.1. MAIN RESULTS: IPCR ACHIEVES 95% OF ORACLE PERFORMANCE

Routing Strategy	Performance	Relative to Oracle
Oracle	91.39%	100.00%
IPCR	86.82%	95.00%
Uniform	36.79%	40.26%
Random	14.97%	16.38%

IPCR achieves 95.00% of oracle performance, exceeding the 90% threshold. The improvement over Uniform is statistically significant ($t = 68.99$, $p = 7.05e-224$), confirming that routing provides substantial benefit beyond naive adapter aggregation.

5.0.2. TASK FAMILY SEPARABILITY (H-E0)

Per-family F1 scores: All 9 tested families exceed $F1 = 0.98$, ranging from 0.982 (`strategyqa`) to 1.000 (`sensemaking`).

Near-perfect linear separability ($F1 = 0.995$) demonstrates that MiniLM embeddings encode strong task-

Metric	Value	Threshold	Status
Macro-F1	0.995	≥ 0.75	Pass
Accuracy	0.998	—	—
Baseline F1	0.115	—	—

type signal. The 8.6x improvement over random baseline (0.995 vs 0.115) confirms the signal is genuine.

5.0.3. ADAPTER SELECTION ACCURACY (H-E1)

Metric	Value	Threshold	Status
Top-1 Accuracy	72.67%	$\geq 70\%$	Pass
Top-3 Accuracy	95.78%	$\geq 85\%$	Pass
vs Random	14.8x	$> 1x$	Pass

The high top-3 accuracy (95.78%) indicates that even when top-1 prediction is incorrect, the correct adapter is nearly always in the top 3 candidates.

5.0.4. ROBUSTNESS ANALYSIS (H-M2)

Robustness was tested under paraphrase variations and keyword masking.

Paraphrase Robustness:

Variant	Cosine Mean	Routing Consistency
WordNet synonyms	0.720	72.5%
Embedding-filtered	0.843	79.6%
Combined	0.782	76.1%

Threshold: Cosine ≥ 0.90 . Result: Fail (0.782)

Keyword Masking:

Threshold: Drop $< 10\%$. Result: Fail (44.4%)

The robustness failure reveals that routing depends on lexical keywords, not semantic invariants. WordNet synonym substitutions cause embedding drift beyond routing tolerance.

Per-Class Analysis:

Tasks with distinctive vocabulary (math: "calculate", "equation") maintain routing stability. Open-domain tasks with less keyword anchoring are most vulnerable.

Masking Rate	Original Acc	Masked Acc	Drop
20% keywords	74.9%	48.9%	26.0%
50% keywords	74.9%	30.4%	44.4%

Task Family	Routing Consistency	Interpretation
cot_qasc	97.2%	Robust — distinctive vocabulary
stream_aqua	89.0%	Robust — math keywords
cot_esnli	49.2%	Fragile — open-domain NLI
cot_creak	62.9%	Fragile — claim verification

6. Discussion

6.0.1. KEY FINDINGS

IPCR achieves 95% of oracle performance using frozen embeddings and a linear probe. This validates that instruction-adapter alignment exists intrinsically without joint training or validation examples.

However, robustness results temper this success. Routing depends on lexical features, not semantic understanding.

What works: FLAN-style instructions with consistent templates and task-indicative keywords. The linear separability ($F1 = 0.995$) and high top-3 coverage (95.78%) make IPCR viable for controlled instruction interfaces.

What does not work: Paraphrased queries, user-generated instructions with non-standard wording, or any setting where keyword anchoring is unreliable.

6.0.2. LIMITATIONS

L1: Lexical Dependence. The 44.4% accuracy drop under keyword masking indicates routing is driven by surface tokens. This is fundamental to MiniLM’s mean-pooling architecture. Mitigation requires either paraphrase augmentation during probe training or switching to encoders with stronger compositional semantics.

L2: Paraphrase Fragility. Cosine similarity of 0.782 under paraphrase means approximately 24% of instruction variations will cause routing inconsistency. Production deployment would require instruction normalization or hybrid routing combining embedding scores with keyword-based confidence.

L3: Task Family Coverage. Testing covered 9-18 task families, not the full 62 FLAN categories. Generalization to untested families is unverified.

L4: Oracle Approximation. H-E1 used task names as oracle proxy rather than computing per-adapter loss for each sample. This is a valid upper-bound estimate given H-E0’s 99.5% task separability, but true oracle

accuracy may differ.

6.0.3. RELATION TO COMPETING EXPLANATIONS

Results are consistent with two interpretations:

1. **Intrinsic alignment hypothesis:** Instruction semantics and adapter specializations genuinely share geometric structure inherited from base model instruction-tuning.
1. **Lexical anchoring hypothesis:** High performance reflects keyword preservation in FLAN templates, not semantic understanding.

The robustness failure (H-M2) provides stronger support for the lexical anchoring interpretation.

7. Conclusion

This work demonstrates that the validation data requirement for adapter routing can be eliminated—instruction prefixes alone contain sufficient signal for effective routing.

IPCR, a minimal approach using frozen MiniLM embeddings and a linear probe, achieves 95% of oracle performance on held-out task families. Instruction embeddings are linearly separable by task family with macro-F1 = 0.995. However, the mechanism depends on task-indicative keywords rather than semantic invariants: paraphrase perturbations cause 24% routing inconsistency, and keyword masking causes 44.4% accuracy drops.

This characterization defines the boundary conditions for IPCR deployment. The method is appropriate for controlled instruction interfaces but requires robustification for open-ended queries.

Future directions include paraphrase augmentation during probe training, testing encoders with stronger compositional semantics (E5-large, Instructor-XL), and hybrid routing combining embedding-based routing with keyword confidence.

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